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Case 19-2026: A 68-Year-Old Man with Fatigue, Fever, and Hypoxemia

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PRESENTATION OF CASE

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CME



Dr. Carlos Portales (Medicine): A 68-year-old man was evaluated in an outpatient clinic of this hospital because of fatigue and dyspnea.

The patient had a history of chronic obstructive pulmonary disease with progressive bronchitis symptoms that had resulted in frequent emergency department visits and hospital admissions, as well as the use of supplemental oxygen during exertion. Three years before the current presentation, bronchoscopic lung-volume reduction surgery was performed, with placement of four endobronchial valves. One year later, symptoms progressed, and the patient was referred to the lung transplantation center of this hospital.

Fifteen months before the current presentation, the patient underwent transplantation of both lungs, with wedge resections of the right middle lobe and left upper lobe. Four months before transplantation, an interferon- γ release assay for the detection of previous exposure to *Mycobacterium tuberculosis* was negative. Donor and recipient serologic testing was performed; the donor was positive for cytomegalovirus (CMV) IgG and Epstein-Barr virus (EBV) IgG, and the recipient was negative for CMV IgG and positive for EBV IgG. At the time of lung procurement, a single granuloma in the left lower lobe was noted incidentally on chest imaging in the donor. In the recipient, histologic examination of the explanted lung tissue showed severe bullous emphysema, reactive lymph nodes, focal necrotizing granulomatous inflammation, and yeast forms. After transplantation, infection prophylaxis with trimethoprim-sulfamethoxazole, ganciclovir, and isavuconazole was started.

The hospital course after transplantation, spanning 8 weeks, was notable for recurrent hypercarbic respiratory failure that led to reintubations and tracheostomy, as well as recurrent aspiration, *Stenotrophomonas maltophilia* pneumonia, drainage of loculated pleural effusions, placement of a gastrojejunostomy tube for nutritional support, acute kidney injury, and encephalopathy. Before discharge, trimethoprim-sulfamethoxazole was discontinued owing to hyperkalemia, and atovaquone was started for *Pneumocystis jirovecii* prophylaxis.

Five months after transplantation, the patient was admitted to this hospital with severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection and was suc-

cessfully treated with remdesivir. Six months after transplantation, prophylaxis with isavuconazole was stopped in accordance with the institutional protocol. Eight months after transplantation, the patient had no respiratory symptoms and was walking and playing golf without the use of supplemental oxygen; the gastrojejunostomy tube was removed.

One year after transplantation, a transthoracic echocardiogram was normal, and routine surveillance bronchoscopy showed patent bronchial anastomoses and no mucosal abnormalities. Histopathological evaluation of transbronchial biopsy specimens showed no features suggestive of allograft rejection; rare fungal organisms suggestive of candida species were noted on methenamine silver staining.

Two months later, and 1 month before the current presentation, the patient presented to an emergency department affiliated with this hospital because of 5 days of fever with a temperature of up to 37.8°C, fatigue, chills, chest tightness, and dyspnea with mild activity; the oxygen saturation measured at home ranged from 90 to 92% while the patient was breathing ambient air at rest and decreased to 85% with ambulation. In the emergency department, the oral temperature was 37.1°C, the heart rate 111 beats per minute, the respiratory rate 22 breaths per minute, and the oxygen saturation 95% while the patient was breathing ambient air. Blood electrolyte levels were normal; other laboratory test results are shown in Table 1. Nucleic acid testing for SARS-CoV-2, influenza virus types A and B, and respiratory syncytial

Table 1. Laboratory Data.*

Variable	Reference Range, Adults†	1 Mo before Current Evaluation, Emergency Department	2 Wk before Current Evaluation, Outpatient Clinic	On Current Evaluation, Outpatient Clinic
Tacrolimus (μg/liter)	Variable; goal of 7.0 to 9.0 in this patient in this time frame	8.7	9.1	9.5
Urea nitrogen (mg/dl)	8–25	26	29	33
Creatinine (mg/dl)	0.60–1.50	1.58	1.50	1.60
Hemoglobin (g/dl)	13.5–17.5	8.9	9.9	9.1
Hematocrit (%)	41.0–53.0	26.9	29.0	28.4
Red-cell count (per μl)	4,500,000–5,900,000	2,190,000	2,490,000	2,400,000
Mean corpuscular volume (fl)	80.0–100.0	122.5	116.5	118.3
White-cell count (per μl)	4500–11,000	4600	6000	7100
Differential count (per μl)				
Neutrophils	1800–7700	—	4270	5370
Lymphocytes	1000–4800	—	860	680
Monocytes	200–1200	—	690	820
Eosinophils	0–900	—	140	60
Immature granulocytes	0–100	—	—	150
Platelet count (per μl)	150,000–400,000	190,000	224,000	203,000
High-sensitivity troponin I (pg/liter)	0–76	12	—	—
D-Dimer (μg/ml)	<0.50	2.12	1.86	—
B-type natriuretic peptide (pg/ml)	<100	—	169	—
Cytomegalovirus DNA (IU/ml)	Negative	—	Detected but too low to be quantified	310
IgG (mg/liter)	614–1295	—	—	588

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

virus was negative. An electrocardiogram and a chest radiograph were reportedly normal.

Dr. Harikrishnan Nandakumar: Computed tomography (CT) of the chest (Fig. 1A and 1B), performed after the administration of intravenous contrast material, showed findings that ruled out pulmonary embolism. A previously identified granuloma in the left lower lobe was present, and a new solid nodular opacity in the right lower lobe was noted.

Dr. Portales: Oral ciprofloxacin was prescribed. Two weeks before the current presentation, repeat laboratory testing showed normal blood levels of electrolytes, albumin, aspartate aminotransferase, and alanine aminotransferase; other laboratory test results are shown in Table 1. CMV DNA was detected in the blood but at a level too low to be quantified. EBV DNA was not detected in the blood.

Dr. Nandakumar: A chest radiograph (Fig. 1C) showed new micronodules and patchy opacities in both lower lobes, predominantly on the right side. CT of the chest (Fig. 1D and 1E) showed extensive new multilobar tree-in-bud and ground-glass opacities that predominantly involved the right lower lobe. As compared with CT images obtained 2 weeks earlier, this study showed an increase in reactive lymph nodes in the hilar, subcarinal, and lower paratracheal regions on both sides. The previously noted solid nodular opacity in the right lower lobe remained unchanged.

Dr. Portales: Spirometry showed a forced expiratory volume in 1 second (FEV₁) of 2.75 liters (90% of the predicted value) and a forced vital capacity (FVC) of 2.81 liters (69% of the predicted value). These findings were consistent with a mild restrictive ventilatory deficit and were similar to the patient's best spirometry results after transplantation.

Two weeks later, the patient returned to the outpatient transplantation clinic of this hospital. He reported fever with a temperature of up to 37.7°C, malaise, morning chills, productive cough with clear sputum, and intermittent dyspnea and chest tightness. The oxygen saturation measured at home ranged from 92 to 94% while he was breathing ambient air at rest. He reported that dyspnea had occurred when he walked from the car to the clinic.

The patient had a history of pulmonary hypertension, systemic hypertension, hyperlipidemia, coronary artery disease (treated with percutaneous

coronary intervention), allergic rhinitis, hiatal hernia, gastroesophageal reflux disease, Schatzki ring (treated with esophageal dilation), tubular adenoma, osteoporosis, and prostatic hypertrophy. Medications included mycophenolate sodium, prednisone, tacrolimus, aspirin, azithromycin (administered in accordance with the institutional protocol), calcium citrate, atovaquone, metoprolamide, metoprolol, mirtazapine, a multivitamin, omeprazole, rosuvastatin, and valacyclovir. Amlodipine had caused edema, and lisinopril had caused a cough.

The patient lived with his wife in a small town in New England. He had retired from work in the heavy machinery industry, where he had been exposed to asbestos, paint, dirt, dust, fumes, and birds. In retirement, he enjoyed machinery restoration and gardening. The patient had smoked one pack of cigarettes daily for 40 years and had quit smoking 12 years before transplantation. He drank one glass of wine weekly and did not use illicit drugs. He had traveled to Florida and the Rocky Mountains but had not traveled outside the United States. The patient was up-to-date with influenza, SARS-CoV-2, and pneumococcal vaccinations. His father had had occupational fume exposures and had died of lung cancer. One sister had chronic obstructive pulmonary disease, and another sister had died of endobronchial cancer in her sixth decade of life. The patient's adult children were healthy.

The patient appeared fatigued. The oral temperature was 36.7°C, the heart rate 109 beats per minute, the blood pressure 126/73 mm Hg, the respiratory rate 16 breaths per minute, and the

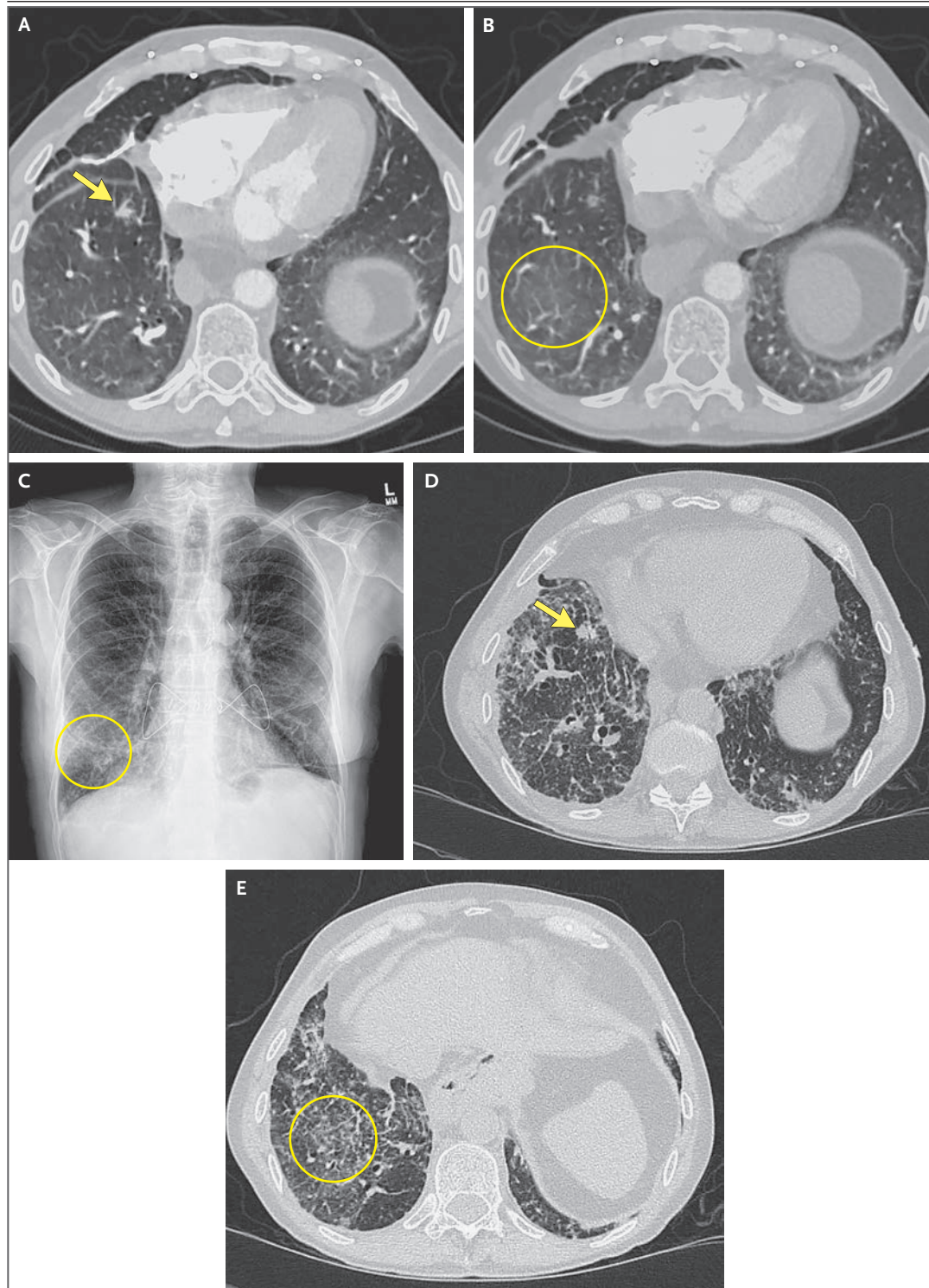
Figure 1 (facing page). Chest Imaging Studies.

CT images of the chest obtained 1 month before the current presentation (Panels A and B) show a new, irregular, solid nodular opacity in the right lower lobe (Panel A, arrow), with no tree-in-bud nodules or ground-glass opacities present in the surrounding parenchyma (Panel B, circle). A chest radiograph obtained 2 weeks before the current presentation (Panel C) shows postsurgical changes from a clamshell sternotomy with transplantation of both lungs. New micronodules and patchy opacities are present in both lower lobes, predominantly on the right side (circle). CT images of the chest obtained 2 weeks before the current presentation (Panels D and E) show that the irregular, solid nodular opacity in the right lower lobe remains unchanged. An air bronchogram is present (Panel D, arrow), and extensive new tree-in-bud nodules and ground-glass opacities that predominantly involve the lower lobes are noted (Panel E, circle).

oxygen saturation 92% while he was breathing ambient air. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 18.7. The sites of the clamshell chest incision and previous tube placement were well

healed, without erythema or drainage. The remainder of the examination was normal.

The blood levels of electrolytes, iron, ferritin, glycated hemoglobin, and methemoglobin were normal; other laboratory test results are shown



in Table 1. The blood level of CMV DNA was 310 IU per milliliter. Cryptococcal antigen testing and an extended respiratory panel — including tests for chlamydia, legionella, mycoplasma infection, and infection with viral pathogens such as adenovirus, influenza virus, metapneumovirus, parainfluenza viruses, respiratory syncytial virus, and human rhinoviruses and enteroviruses — were negative.

Dr. Nandakumar: CT of the chest, performed with the administration of intravenous contrast material, showed a nodule in the right lower lobe that measured 1 cm in diameter, multiple new nodules in the middle and lower lobes that measured 2 to 3 mm in diameter, and an increase in mediastinal lymphadenopathy from 1 month earlier.

Dr. Portales: Four days later, the patient returned to the outpatient clinic to undergo bronchoscopy with transbronchial biopsy. Additional results of tests performed during the previous visit had become available: coccidioides antibody testing was negative, and the serum level of donor-derived cell-free DNA was 0.57%, near the patient's baseline level. Before the procedure, the oxygen saturation ranged from 87 to 90% while the patient was breathing ambient air at rest. Bronchoalveolar-lavage (BAL) fluid was colorless, with 5 neutrophils per microliter. Gram's staining of BAL fluid revealed rare neutrophils and no organisms; a sample was sent for culture. Preliminary histologic examination showed patchy, mixed histiocytic and chronic inflammatory infiltrates within alveolar walls.

Diagnostic test results were received, and management decisions were made.

DIFFERENTIAL DIAGNOSIS

Dr. Hilary J. Goldberg: This 68-year-old man underwent transplantation of both lungs because of chronic respiratory failure in the context of severe chronic obstructive pulmonary disease. After a complicated postoperative course, the patient had excellent functional status by 1 year after transplantation. However, 14 months after transplantation, he had a subacute onset of low-grade fever, chills, fatigue, cough, and exertional dyspnea, symptoms that persisted for 1 month. He currently presents with tachycardia, hypoxemia at rest, macrocytic anemia, lymphopenia, low-level CMV viremia, new nodules in both lungs,

and mediastinal lymphadenopathy on chest imaging. In building a differential diagnosis, I will consider the timing of this patient's symptom onset relative to lung transplantation, in addition to his immunocompromised state and epidemiologic exposures.

ALLOGRAFT REJECTION

Acute cellular rejection, which can cause fever, malaise, cough, shortness of breath, and allograft dysfunction, most commonly occurs in the first year after transplantation. Some features of acute cellular rejection, including fever, dyspnea, and productive cough, can mimic infection. Symptomatic acute cellular rejection is typically associated with a decline in spirometry values (specifically, the forced expiratory flow between 25% and 75% of the FVC, as well as the FEV₁). Although this patient had many clinical features that were suggestive of allograft rejection, the findings on multiple transbronchial biopsies were not consistent with this diagnosis, which is typically characterized by a lymphocytic infiltrate affecting the terminal bronchioles and perialveolar capillaries.¹ Furthermore, the negative serum donor-derived cell-free DNA test also argues against a diagnosis of acute cellular rejection.

CHEMICAL PNEUMONITIS DUE TO ASPIRATION

This patient has risk factors for aspiration, including a hiatal hernia and gastroesophageal reflux disease. He also had recurrent episodes of aspiration during his initial post-transplantation course. However, chemical pneumonitis due to aspiration typically has a more acute onset than the illness seen in this patient. In addition, no foreign material was noted on histopathological examination of the transbronchial biopsy specimens.

BACTERIAL INFECTION

This patient's subacute illness and imaging findings, including a nodular infiltrate, are not suggestive of a typical acute bacterial infection such as community-acquired pneumonia. *Nocardia* is a partially acid-fast, gram-positive rod that can cause opportunistic infections in lung-transplant recipients. Lung nodules on chest imaging are a common finding. However, Gram's staining of the BAL specimen in this patient was negative, a finding that makes community-acquired bacterial infection and *nocardia* infection unlikely.

CYTOMEGALOVIRUS INFECTION

Lung-transplant recipients are susceptible to respiratory viral infections at any time after transplantation. However, nucleic acid testing for common respiratory viruses is a moderately sensitive diagnostic strategy and was negative in this patient. Other viruses to consider include CMV, which can cause CMV syndrome with associated damage to almost any organ system. In immunocompromised hosts, several organ systems are particularly vulnerable to injury: the eye (retinitis and vision loss), the gastrointestinal tract (esophagitis, gastritis, colitis, and hepatitis), the central nervous system (encephalitis, myelitis, and cerebellitis), the lungs (pneumonitis), the bone marrow (pancytopenia), the liver and spleen (hepatitis and splenomegaly), and the kidneys (nephritis and acute kidney injury). However, this patient's imaging studies were not suggestive of pneumonitis, which would also be unusual in the absence of high-level CMV viremia.

EBSTEIN–BARR VIRUS INFECTION

EBV-associated post-transplantation lymphoproliferative disorder (PTLD) is most likely to occur when an organ from a donor with previous EBV exposure (positive for EBV IgG) is donated to a recipient without previous EBV exposure (negative for EBV IgG). Patients with PTLD can present with a nodular pulmonary infiltrate, which was seen in this patient. The presence of EBV viremia can support the diagnosis. However, isolated lung involvement tends to occur earlier after transplantation. In addition, the preliminary pathological findings in this case were not typical of this disease.

MYCOBACTERIAL INFECTION

Although *M. tuberculosis* infection is not more common in lung-transplant recipients than in the general population, reactivation of latent *M. tuberculosis* infection is more common in patients with immunosuppression. However, this patient had a negative interferon- γ release assay before transplantation, which makes reactivation of latent tuberculosis highly unlikely.

Atypical mycobacteria — including the *M. avium* complex and other rapidly growing mycobacteria, such as *M. abscessus*, *M. fortuitum*, and *M. chelonae* — are widely recognized as pathogens that frequently lead to infection in lung-transplant recipients. These organisms often colonize the airway

before transplantation and may cause recrudescence after transplantation. In such cases, evidence of infection is often seen in explanted lung tissue. This patient's explanted lung tissue showed granulomatous inflammation but no acid-fast bacilli on histologic examination. Furthermore, the absence of evidence of mycobacteria on multiple lung biopsies after transplantation and the patient's clinical course make active atypical mycobacterial infection an unlikely cause of his current illness.

FUNGAL INFECTION

Fungal infections are an important consideration in this patient. Previous SARS-CoV-2 infection may have conferred a predisposition to superinfections or reactivation of latent infections.² His subacute illness, which was accompanied by laboratory abnormalities such as macrocytic anemia, also suggests a nonbacterial cause. Lymphadenopathy and nodular and ground-glass opacities on imaging suggest a fungal infection, as does the finding of histiocytes on preliminary pathological analysis.

ASPERGILLOSIS

Infection with aspergillus species is the most common type of fungal infection in lung-transplant recipients. Patients with aspergillosis can present with nodular disease, which was seen in this patient. However, aspergillosis in lung-transplant recipients is often associated with airway abnormalities, such as friability, erythema, plaques, or stenosis. Invasive aspergillosis can also lead to hemoptysis. An elevated galactomannan level in BAL fluid or serum can support the diagnosis, although the absence of this finding does not rule out the disease. The absence of branching hyphae that are consistent with aspergillus in the biopsy specimens does not support this diagnosis.

CRYPTOCOCCAL INFECTION

Cryptococcus is an opportunistic fungus that can infect the lungs, brain, and other organs. Cryptococcus appears as a mucoid yeast, and cryptococcal antigen testing was negative in this patient, which can help rule out the diagnosis. The absence of extrapulmonary disease argues against cryptococcosis, and the findings on multiple biopsies were not consistent with this diagnosis.

PNEUMOCYSTIS JIROVECIi INFECTION

Lung-transplant recipients are susceptible to *P. jirovecii* infection at any point after transplantation, although the risk decreases over time. This patient was receiving atovaquone prophylaxis. Although trimethoprim–sulfamethoxazole is the preferred first-line agent for both treating and preventing *P. jirovecii* infection, atovaquone is also effective in preventing *P. jirovecii* pneumonia. In addition, the imaging and histologic findings in this case were not typical for this infection.

ENDEMIC FUNGAL INFECTION

Histoplasmosis is most common in the Ohio River and Mississippi River Valleys, but it occasionally occurs in New England. This patient's gardening activity and previous exposure to birds place him at risk for exposure to *Histoplasma capsulatum*, a dimorphic fungus commonly found in soil.³ On histopathological examination, the pathogen appears as a small, narrow-based budding yeast. The morphologic appearance can cause it to be mistaken for candida on methenamine silver staining, especially when the yeast forms are small or poorly visualized. The yeast forms seen on biopsy in this patient 1 year after transplantation that were thought to be candida were more likely to be *H. capsulatum*.

Blastomycosis and coccidioidomycosis can have clinical features similar to those of *H. capsulatum* infection,⁴ but both are uncommon in New England. Blastomyces species appear larger than *H. capsulatum* on tissue sampling. Coccidioides appears as spherules, which were not identified in any biopsy specimens in this patient. In addition, coccidioides antibody testing was negative.

Histoplasmosis can lead to asymptomatic lymphadenopathy or incidental pulmonary nodules, as well as to cavitary lung lesions or diffuse pulmonary infiltrates with associated cough, dyspnea, and constitutional symptoms.⁵ Donor-derived histoplasmosis typically develops in the initial months after transplantation,⁶ whereas reactivation of latent infection in the recipient usually occurs months to years later. Although imaging in the donor at the time of organ procurement showed an isolated granuloma in the left lower lobe, the disease course in the recipient is more consistent with reactivation of latent disease; however, it is not possible to determine whether this infection was reactivation of latent disease in the recipient or was donor-derived. The presence of necrotizing granulomatous inflammation with

yeast forms in this patient's explanted lung tissue further supports reactivation of histoplasmosis. Antigen testing of blood or urine can support the diagnosis, but I suspect that the diagnostic test in this patient was a fungal culture of the BAL specimen.

DR. HILARY J. GOLDBERG'S
DIAGNOSIS

Histoplasmosis.

DIAGNOSTIC TESTING

Dr. Lida P. Hariri: Histopathological examination of a transbronchial biopsy specimen (Fig. 2), obtained in an assessment for allograft rejection, showed patchy, mixed histiocytic and chronic inflammatory infiltrates within the alveolar walls. These findings were suggestive of infection. Special staining for organisms, including staining for acid-fast bacilli, Grocott–methenamine silver staining, and Brown–Hopps staining, was negative. However, the absence of organisms on special staining does not rule out infection; the sensitivity of the staining is low. Aspiration was also included in the differential diagnosis but was considered to be unlikely, given the histopathological findings. There was no evidence of allograft rejection.

Dr. Briana N. Castillo: In this patient, serum and urinary antigen enzyme immunoassays detected histoplasma and blastomyces species. Complement fixation and immunodiffusion antibody testing for histoplasma were negative. However, a definitive diagnosis relies on identifying the pathogen on histopathological examination, cytopathological examination, or culture. Serologic and antigen-based test results can support a presumptive diagnosis but are neither confirmatory nor sufficient to rule out histoplasmosis.⁷

Fungal culture remains the standard method for the diagnosis of histoplasmosis. *H. capsulatum* is a thermally dimorphic fungus, growing as a mold at 25 to 30°C and as a yeast at 37°C. Mycelial colonies appear white or brown with a dense, cottony texture. Growth in culture can take 2 to 8 weeks. In this case, fungal culture of the BAL specimen was notable for growth at 21 days of incubation (Fig. 3A). Examination of the mold colony with lactophenol cotton blue staining showed septate hyphae, pear-shaped microconidia, and tuberculate macroconidia that are

characteristic of *H. capsulatum* (Fig. 3B). Because the macroconidia of *H. capsulatum* are morphologically similar to those of *sepedonium* species, identification was confirmed by means of

matrix-assisted laser desorption ionization–time-of-flight (MALDI-TOF) mass spectrometry.⁸ In addition, cytologic examination of the BAL specimen with Grocott–methenamine silver staining revealed intracellular, narrow-based budding yeasts that measured 2 to 4 μm in diameter, with morphologic features consistent with *H. capsulatum* (Fig. 3C).

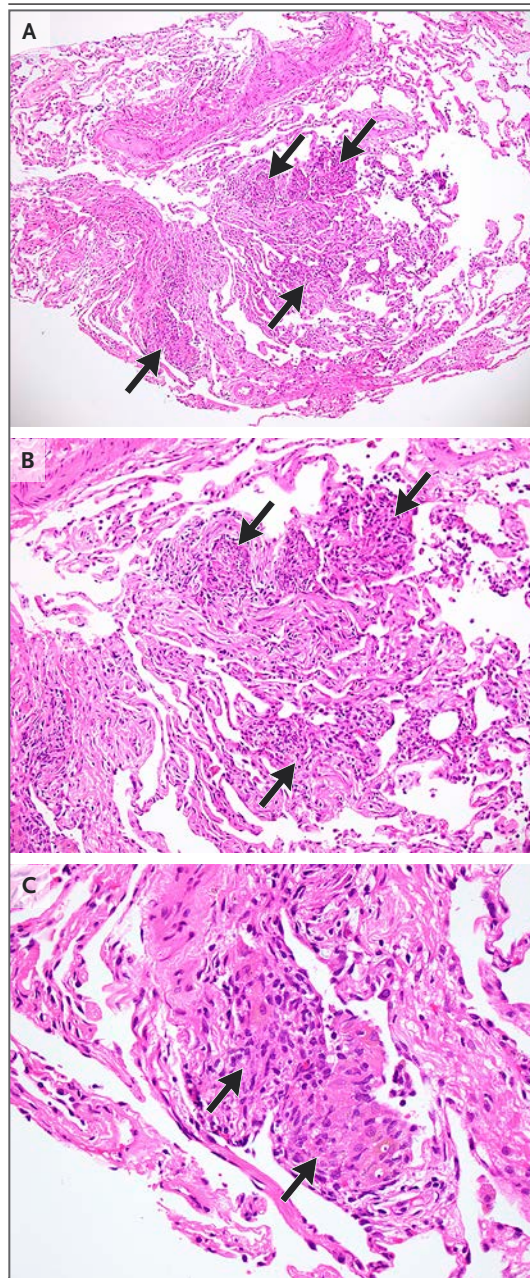


Figure 2. Lung-Biopsy Specimen.

Hematoxylin and eosin staining of a transbronchial biopsy specimen, obtained in an assessment for allograft rejection, is shown at various degrees of magnification (Panels A, B, and C). Patchy, mixed histiocytic and chronic inflammatory infiltrates are present within the alveolar walls (arrows), findings suggestive of infection. There is no evidence of allograft rejection.

MICROBIOLOGIC DIAGNOSIS

Histoplasmosis.

DISCUSSION OF MANAGEMENT

Dr. Arthur Y. Kim: When we evaluated this patient, he had no evidence of disseminated histoplasmosis outside the lungs (e.g., involvement of the skin, bone marrow, central nervous system, or adrenal glands). Nevertheless, the possibility of occult dissemination was high, given the depth and duration of his immunosuppression. This possibility was supported by the diagnosis of low-level CMV viremia; multiple simultaneous infectious processes in solid-organ transplant recipients are common. Treatment with valganciclovir was initiated.

For antifungal therapy, treatment with liposomal amphotericin B was started. A course of liposomal amphotericin B was to be administered for 1 to 2 weeks, followed by prolonged oral therapy.⁹ The day after administration of the first dose, the patient had a temperature of 38.7°C, and after the second dose, he noted a marked decrease in his malaise and fatigue. The creatinine level increased to 2.57 mg per deciliter (227 μmol per liter), with the change attributable to amphotericin therapy. In a phase 2 randomized, controlled trial, outcomes with a single dose of liposomal amphotericin B were similar to those with 2 weeks of daily therapy with liposomal amphotericin B among persons with advanced human immunodeficiency virus infection and disseminated histoplasmosis.¹⁰ Extrapolating from these data and the potential consequences of further kidney injury, we elected to discontinue amphotericin therapy and start oral therapy. Although many physicians would prescribe itraconazole for this condition, posaconazole was chosen owing to more reliable absorption and manageable drug interactions.¹¹ The patient's acute kidney injury resolved.

Our team also considered the source of exposure to prevent future reinfection. *H. capsulatum*

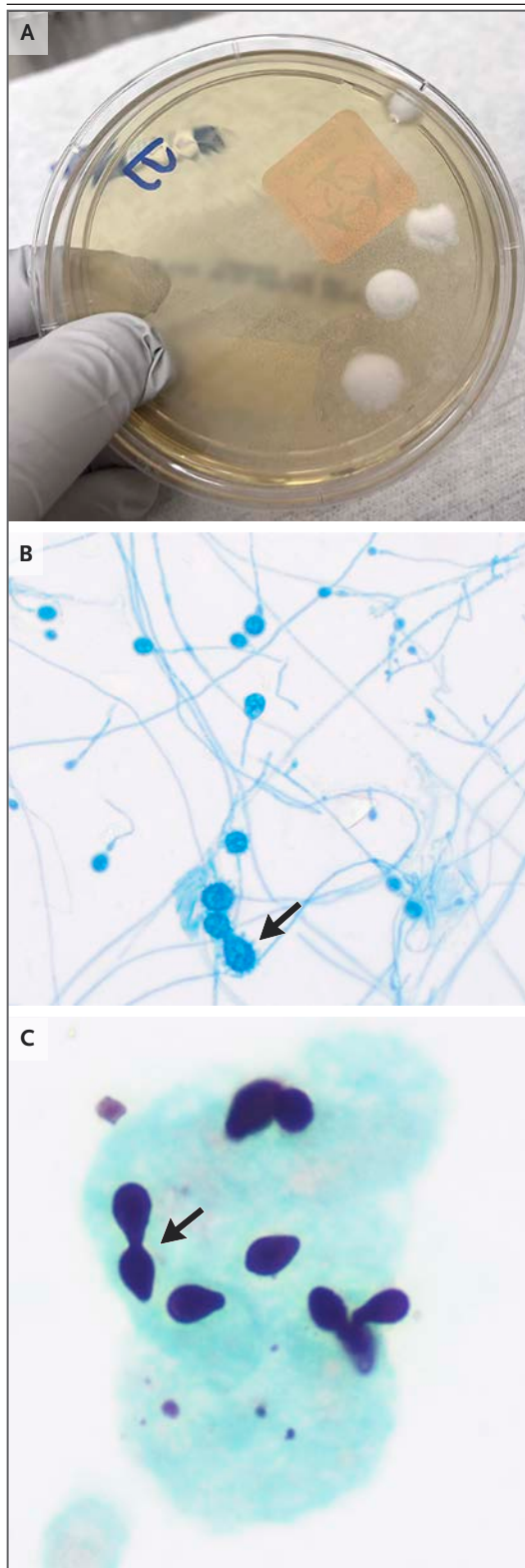


Figure 3. Bronchoalveolar-Lavage Specimen.

After 21 days of incubation at 30°C, Sabouraud dextrose agar shows the growth of white, cottony mold colonies that are consistent with *Histoplasma capsulatum* (Panel A). Lactophenol cotton blue staining shows septate hyphae with tuberculate macroconidia that are characteristic of *H. capsulatum* (Panel B, arrow). Cytologic examination with Grocott-methenamine silver staining shows numerous small, oval, narrow-based budding yeasts within macrophages (Panel C, arrow).

enters the host after inhalation of spores found in soil contaminated by bird or bat droppings. In the United States, histoplasmosis is most common in the Midwest; however, we have seen cases in the Northeast, which reflects an increased risk outside regions where the disease is traditionally endemic.¹² In the context of solid-organ transplantation, we considered the possibility of donor-derived infection and determined that this scenario was unlikely because most cases develop in the initial months after transplantation, at a time of maximal immunosuppression, and because this patient had received prolonged antifungal prophylaxis with activity against histoplasma species, in accordance with the protocol at this institution. To prevent recurrent infection, we recommended indefinite oral antifungal therapy on the basis of the patient's age, the severity of his presentation, and possible ongoing environmental exposures.

Dr. Brian C. Keller: The goal for the tacrolimus trough concentration in this patient had been lowered to 7 to 9 ng per milliliter in accordance with the institutional protocol for patients at 1 year after transplantation; therefore, no further dose adjustments were made after the diagnosis of histoplasmosis. To reduce immunosuppression, mycophenolate sodium therapy was withheld on the patient's admission to the hospital. Treatment with mycophenolate sodium was eventually resumed 5 months later, after the patient had completed antifungal therapy and was receiving lifelong antifungal prophylaxis. During the period of reduced immunosuppression, care was taken to monitor for immune reconstitution inflammatory syndrome, a process characterized by fever, weight loss, fatigue, and shortness of breath.¹³

Triazole antifungal agents have varying degrees of interaction with calcineurin inhibitors, such as tacrolimus, through inhibition of cytochrome P450 3A4.¹⁴ After the initiation of posaconazole therapy, monitoring of tacrolimus and posaconazole drug levels, with frequent adjustment of the tacrolimus dose, was necessary to prevent toxic effects from supratherapeutic tacrolimus levels and to prevent allograft rejection from subtherapeutic levels, neither of which was observed in this patient.

The risk of recurrence of histoplasmosis after treatment is approximately 6%, with most recurrences occurring within 2 years after the diagnosis. Failure to reduce the calcineurin inhibitor dose has been linked to a higher risk of recurrence in solid-organ transplant recipients.¹⁵ Serial monitoring of histoplasma antigen levels is recommended in these patients during treatment and for at least 6 months after completion of therapy.⁹

Dr. Portales: This patient had a good response to antifungal therapy. After 1 week, he was discharged home without the use of supplemental

oxygen. At his last follow-up visit, 4 months after admission, he continued to do well and was walking approximately 1 mile per day.

FINAL DIAGNOSIS

Histoplasmosis.

This case was presented at the Harvard Medical School postgraduate course "Internal Medicine: Comprehensive Review and Update 2025," directed by Jose C. Florez, Rocio M. Hurtado, and David M. Dudzinski.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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